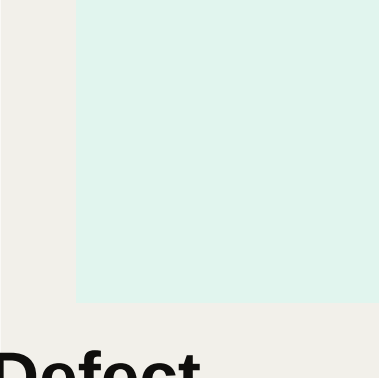




LEAN SIX SIGMA CASE STUDY

Reducing Returns Through Defect Root Cause Analysis

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Reduced warranty returns from 24 to a target of 5/month — identifying galvanometer calibration as root cause of 54% of all field failures and delivering \$161,280 in annual labour savings.

TOOLS & METHODS

DMAIC Framework · Problem & Objective Statements · SIPOC · Pareto Analysis · 5-Why Root Cause · Cost-Benefit Analysis · Process Mapping

Executive Summary

A Time-of-Flight Laser Inspection System manufacturer was losing 58% of available working hours daily to warranty returns and rework — directly compressing capacity for new order production. This LSS Define-phase project quantified the financial impact, mapped the return loop, identified the vital few defect categories through Pareto analysis, and traced the dominant root cause (galvanometer calibration failure) to a systemic gap in the final test protocol.

Business Problem

Industry	Precision Optical / Laser Systems Manufacturing
Product	Time-of-Flight Laser Inspection Systems — installed at customer sites
Warranty Period	1 year post-installation — all returns covered; 30 replacement units held in reserve
Problem	288 customer returns/year (24/month) consuming 280 hours/month of skilled technician time for troubleshooting and rework. 58% of daily working hours lost to warranty work — directly reducing new order capacity.
Financial Impact	\$16,800/month rework labour cost at \$60/hr blended rate
Target	Reduce returns from 24 to 5/month (80% reduction) by January 2024 — saving \$161,280/year

Objectives

- Quantify the full financial cost of the current return and rework burden.
- Identify the vital few defect categories driving the majority of returns (Pareto 80/20).
- Trace the dominant defect to its root cause using structured 5-Why analysis.
- Define immediate and short-term corrective actions with measurable KPIs.

APPROACH

Methodology — DMAIC Framework

Phase	Activity	Tool / Output
Define	Scoped problem: 24 returns/month consuming 58% of capacity Defined objective: 80% reduction = 5 returns/month Mapped full warranty return loop (RMA → repair → replacement pool)	Problem Statement Objective Statement Process Map
Measure	Collected October 2023 defect frequency data (n=24 returns) Categorised all returns by defect type Calculated baseline sigma level from defect rate Quantified labour hours and cost per return	Defect Frequency Table Baseline Cost Analysis \$16,800/month
Analyze	Pareto analysis: identified top 3 defects = 79% of all returns Galvanometer defects (both axes combined) = 54% of all returns 5-Why analysis on top defect: traced to missing final test gate Root cause: no FMEA performed at product launch	Pareto Chart 5-Why Analysis Process Gap Identification
Improve	Developed 6 recommendations across immediate / short-term / ongoing Prioritised by impact: final test protocol update eliminates 33% immediately FMEA mandate for future products prevents recurrence	Corrective Action Plan KPI Targets
Control	Monthly Pareto dashboard for ongoing return tracking CAPATrigger: any single defect class >2 units/month Replacement pool reduction from 30→10 once target achieved	Control Plan KPI Dashboard

Cost-Benefit Calculation

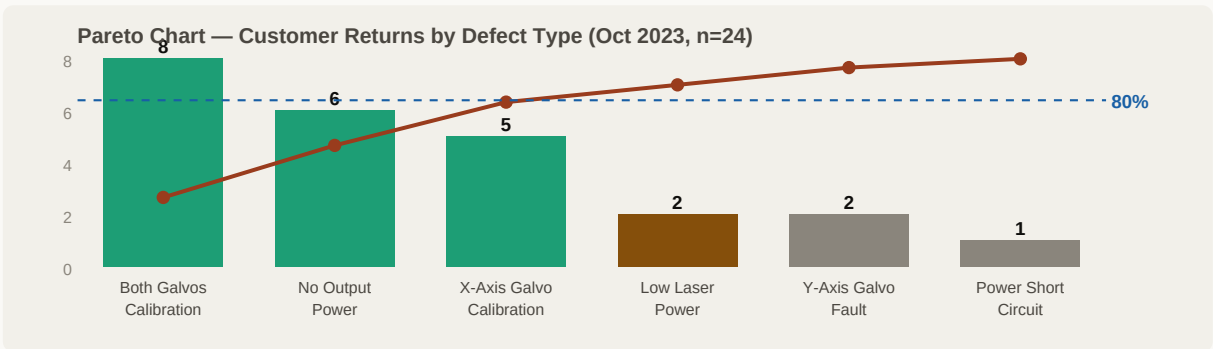
Metric	Baseline (Oct 2021)	Post-Improvement	Saving
Returns/month	24 issues	5 issues	80% reduction
Rework hours/month	280 hrs	56 hrs	224 hrs freed
Labour cost/month	\$16,800	\$3,360	\$13,440/month
Annual labour cost	\$201,600	\$40,320	\$161,280/year
Replacement pool	30 units (tied capital)	10 units	20 units freed

RESULTS

Analysis & Key Findings

24/mo	58%	79.1%	\$161K
Baseline Returns	Hours Lost to Rework	Top 3 Defects Share	Annual Savings Target

Pareto Analysis — Defect Frequency (October 2023, n=24 returns)



The top 2 defect categories — Both Galvos Out of Calibration (33.3%) and No Output Power (25.0%) — account for 58.3% of all returns. Galvanometer-related defects across all three categories (Both, X-axis, Y-axis) represent 54% of total returns. Addressing calibration alone nearly achieves the 80% reduction target.

5-Why Root Cause Analysis — Primary Defect: Galvo Calibration

5-Why Root Cause Analysis — Both Galvos Out of Calibration (33% of all returns)

Why 1:	Calibration not verified in final test & close process before shipment	Occurrence
Why 2:	Final test protocol had no mandatory dual-galvo calibration check step	Occurrence
Why 3:	Test protocol written before galvo calibration drift identified as field failure	System
Why 4:	No structured FMEA performed at product launch to identify failure modes	System
Why 5:	Engineering quality process did not mandate FMEA as a design release gate	Root Cause

Root cause confirmed: the absence of a mandatory FMEA at product launch meant galvanometer calibration drift was never identified as a failure mode — and therefore no control was ever built into the final test protocol. This is a systemic process design gap, not an isolated operator error.

Recommendations

Priority	Action	KPI Target
IMMEDIATE	Update Final Test Protocol — add mandatory dual-galvo calibration verification step. Minimum threshold: $\pm 0.01^\circ$ alignment accuracy. Technician sign-off required before unit close.	0 galvo-related returns within 90 days
IMMEDIATE	Add Power Output Pre-Ship Check — minimum output power threshold test ($\geq$ rated watts $\pm 5\%$) as final inspection step. Eliminates defect class #2 = 25% of returns.	Power returns $\rightarrow$ 0 within 60 days
SHORT-TERM	Conduct Retrospective FMEA — root cause traced to FMEA absence at launch. Complete FMEA on current product family; mandate FMEA as release gate for future products.	FMEA complete within 30 days
SHORT-TERM	Technician Training on Galvo Alignment — structured training module on calibration theory, procedure, and acceptance criteria. Certify all bench technicians before solo sign-off.	100% certification within 45 days
ONGOING	Monthly Return Rate KPI Dashboard — Pareto tracking by defect category. CAPA trigger if any single class exceeds 2 units/month.	Dashboard live within 2 weeks
ONGOING	Reduce Replacement Pool 30 $\rightarrow$ 10 units — once returns reach ≤ 5 /month target, release 20 units of tied capital. Revisit pool size quarterly.	Pool optimised within 6 months

Financial Impact & Expected Benefits

\$161,280	224 hrs	80%	20 units
Annual Labour Savings	Monthly Capacity Freed	Return Rate Reduction	Capital Released (Pool)

Benefit Category	Current State	Improved State	Value
Labour cost (rework)	\$16,800/month	\$3,360/month	\$161,280/year saved
Production capacity	58% hours lost to rework	<12% hours on returns	224 hrs/month reclaimed
Replacement pool	30 units held in reserve	10 units sufficient	20 units freed (~\$X capital)
Customer satisfaction	24 complaints/month	≤5 complaints/month	NPS and contract protection
Engineering process	No FMEA — reactive	FMEA at every launch — proactive	Prevents recurrence on future products

- Reclaiming 224 hours/month of skilled technician time is equivalent to re-activating a full production resource — enabling new orders to be fulfilled without additional headcount.
- The root cause fix (updating the final test protocol) is a zero-cost corrective action — it requires only an engineering document change and a 30-minute technician briefing, yet it directly eliminates 33% of all returns.
- Mandating FMEA as a product release gate is an investment that pays across the entire future product portfolio, not just this product family.

Key Insights & Future Opportunities

Key Insights

- **Pareto 80/20 accelerates focus:** with 6 distinct defect categories, it would be tempting to try to fix all 6. Pareto analysis showed that focusing only on galvanometer calibration — one issue in one test protocol step — eliminates 54% of all returns. Prioritisation is a force multiplier.
- **5-Why exposes system failures, not people failures:** the initial assumption was that technicians were not calibrating correctly. The 5-Why revealed the real problem was that no one had ever been asked to — the test protocol simply didn't include it. Blame the process, not the person.
- **The cost of no FMEA is quantifiable:** \$161,280/year in rework cost can be directly traced to the decision not to conduct an FMEA at product launch. FMEA is not a bureaucratic exercise — it is an investment with a calculable return.
- **The warranty return loop is a hidden factory:** 58% of working hours consumed by returns means the real production capacity of the facility was less than half its stated capacity. VSM of the warranty return loop made invisible waste visible.

Future Opportunities

- Extend the Measure phase to calculate a formal baseline Sigma level from defect data — enabling pre/post comparison once improvements are implemented.
- Apply the same Pareto + 5-Why methodology to other product lines in the portfolio to identify whether galvo calibration is a systemic issue across products.
- Design a poka-yoke in the test software that cannot advance to 'Final Close' status unless the galvo calibration verification step is logged with a passing result.

Core Competencies Applied

Skill / Competency	Evidence in This Project
Lean Six Sigma (DMAIC)	End-to-end Define phase; structured problem → objective → data → root cause → action plan
Pareto Analysis	80/20 applied to defect frequency data; vital few identified (top 3 = 79% of returns)
5-Why Root Cause Analysis	5-Why traced galvo calibration failure to systemic FMEA absence at product launch
Cost-Benefit Analysis	Full financial model: \$16,800/month rework cost → \$161,280/year savings quantified

Process Mapping	Warranty return loop mapped; SIPOC-style flow from RMA to replacement pool
Problem & Objective Statements	SMART objective: 24→5 returns/month by Jan 2024; measurable, time-bound, financially grounded